

PAPER_526 — 3D-IPO Non-Linear Three-Helix Progression Overlay

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

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Author: Daniel T. Murphy **Framework:** Star-Magic / UQFF **Version:** v5.02 **Date:** 2026-03-25 **Session:** 142 — grok_share_2515709ed.txt **CP4 Class:** ThreeDIPONonLinearProgressionCalculator (#121) **Quality Score (QS):** 5 / 5

§1 — Overview

The **3D-IPO** (Three-Dimensional Irrational-Progress Overlay) framework models the simultaneous progression of three distinct physical axes as interlocking helices in a 26-dimensional UQFF braid space:

Helix	Axis	Description
H■	Wolfram progression	Computational irreducibility path through state space
H■	π progression	Irrational decimal expansion trajectory
H■	F_U_Bi_i axis	Buoyancy-force magnitude sequence

Because π is irrational, the crossing pattern of H■ and H■ never repeats — generating a **topologically unique braid fingerprint** for every physical system.

§2 — Core Equation

$$n_{\text{cross}} = \arg\min_n |W_{\text{prog}}(n) - \Pi_{\text{prog}}(n) \cdot F_{U_{Bi}}(x)|$$

where:

Symbol	Definition
$W_{\text{prog}}(n)$	Wolfram computation depth at step n
$\Pi_{\text{prog}}(n)$	$\lfloor 10^n \pi \rfloor \bmod 10$ — n -th π digit
$F_{U_{Bi}}(x)$	UQFF buoyancy force at parameter x
n_{cross}	First crossing index (braid primary node)

§3 — UQFF Number Systems Integration (PAPER_429)

Vacuum Density Series (VDS)

$$A_{\text{helix}} = \mathrm{Li}_{26}([SSq]) = \sum_{k=1}^{26} \frac{[SSq]^k}{k^{26}} \approx 0.570$$

VDS normalises each helix amplitude, ensuring all three axes share the same 26-dimensional natural units. The common amplitude anchor prevents artificial scale separation between Wolfram, π , and $F_{U_{Bi}}$ progressions.

Dipole Vortex Primes (DVP)

$$\Delta_{\text{vortex}} = p_{\text{special}} = 113 \quad (p > 26)$$

Prime 113 governs the vortex node spacing on all three helix axes. As the first prime beyond the 26-dimensional scale of UQFF, it encodes the shortest non-reducible interval between physically distinct crossing events.

§4 — Braid Topology Proof

Proposition: The 3D-IPO braid has no repeating sub-word.

Proof sketch:

1. H is driven by π digit sequence, which is conjectured (and computationally verified to 100 trillion digits) to be **normal** — every finite digit string appears with equal frequency but never periodically.
2. H follows Wolfram computation depth, which by the **Principle of Computational Irreducibility** cannot be compressed to a shorter rule.
3. The crossing condition $W_{\text{prog}}(n) = \Pi_{\text{prog}}(n) \cdot F_{U_{Bi}}(x)$ requires simultaneous coincidence in two irreducible sequences $\rightarrow$ probability of periodicity = 0.
- $\boxed{P(\text{braid repeats}) = 0}$

§5 — Available Equations

Equation	Description
$\text{braid}(n) = \sum_{\text{helix}} e^{i\pi n/p_k}$	Phase-space braid representation
$\rho_{\text{cross}} \propto 1/\log(n)$	Non-repeating crossing density
$A_k = \text{Li}_{26}(\text{SSq})$	Helix amplitude from VDS
$\Delta_k = p_{\text{special}} = 113$	Vortex node spacing from DVP

§6 — Observational Implications

- **Galaxy rotation curves:** Each galaxy leaves a unique 3D-IPO fingerprint in its UQFF buoyancy field, distinguishing it from all other systems.
- **Pulsar timing:** Pulsar spin-down sequences correspond to H axis samples; non-repeating braid predicts no exact period doubling.
- **Wolfram Hypergraph:** 3D-IPO crossing events correspond to causal cone intersections in the Wolfram physics hypergraph (SOURCE116).

§7 — CP4 Calculator Output

```
calc = ThreeDIPONonLinearProgressionCalculator()
result = calc.compute(dataset={'SSq': 0.57}, n_steps=1000)
# result[n_cross] - first crossing index
# result[crossing_count] - total crossings in n_steps
# result[braid_topology] - NON_REPEATING (irrational  $\pi$ )
```

§8 — References

- PAPER_429: Three New UQFF Number Systems (VDS / DVP / BH)
- SOURCE116: Wolfram Hypergraph Emergent Spacetime
- grok_share_2515709ed.txt: BigBangHypergraphTheory Millennium proof set
- Wolfram, S. (2020): *A Project to Find the Fundamental Theory of Physics*